Secure Federated Learning in Healthcare Applications: Diabetes Detection

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***Abstract*—Federated learning (FL) presents a unique approach for collaborative decentralized model training for IoT devices ensuring raw data is not exposed during the process. However, FL faces security challenges during data transmission, as aggregated model up- dates exchanged between client nodes and the central server are vulnerable to leakage and tampering. This issue remains a challenge for applications in IoT where both privacy and data integrity are paramount, such as in healthcare, finance, smart cities, and industrial IoT (IIoT). This study addresses these challenges by proposing a novel framework for FL in IoT environments combining Homomorphic Encryption using TenSEAL for confidentiality with HMAC-SHA256 for integrity and authentication. By uniquely using this dual-layered approach, it provides both confidentiality and integrity—optimized for IoT's constraints. Planned experiments will assess the framework's effectiveness while measuring computational and communication overhead, scalability, and performance on constrained IoT devices.**

***Index Terms*—Federated Learning, Internet of Things IoT, IIoT, Homomorphic Encryption (TenSEAL), HMAC-SHA256, Confi- dentiality, Integrity**

I. Introduction

Federated learning (FL) is an emerging approach for remote collaborative model training while preserving sensitive information, improving a global common model iteratively. Each node receives the model from a shared central server (pre-trained foundation model). The raw data on each device is then used to train the model locally, increasing data privacy. Then each node sends back their updated weight, which are then aggregated and used to obtain an improved global model. [1] FL is divided into two formed of collaborative ML, Horizontal and Vertical Federated Learning. Horizontal Federated learning is the approach in which two participants share common objectives with similar datatypes with different user data (e.g: two banks or hospitals), model updates are then shared without exchanging these raw data, preserving data confidentiality. Vertical Federated learning applies when two participants have the same user data but different objectives (e.g: e-commerce platforms and banks). [2]

This decentralized and distributed training process is being utilized in IoT environments, where this approach would revolutionize IoT in different sectors. Since IoT devices continuously collect sensitive and crucial data, these data could be utilized by ML models to learn patterns that were previously unnoticed, identify future gaps, and optimize services. IoT has been rapidly growing during the last few years. The total installed base of IoT-connected devices has increased by almost triple with 11.7 billion connections in 2020 to a projected 30.9 billion connections in 2025. [3] In 2018, IoT was expected to generate more than 500 zettabytes of data per year — when the amount of installed base of IoT-connected devices was 8 billion connections. [4]

The implementation of FL-based IoT applications covers fields such as medicine with IoMT (Internet of Medical Things), IIoT (Industrial Internet of Things), finance, and smart cities— where in each it is required that data is transferred from each node to the central server. [5] However, we face different challenges: 1- Leaked updates communicated between nodes and central servers could be utilized to retrieve underlying private information. 2- Leakage of these parameters open the door for active attacks that allow tampering the data or the aggregated model. These types of attacks leave us with an unchecked threat to the integrity and privacy of these systems [6]

Federated Learning's structure can be utilized by applications in network traffic management, smart cities, and healthcare in the context of IoT. These systems benefit from FL's ability to manage diverse and large data streams and sources while preserving sensitive user information, especially when coupled with encryption techniques responsible for the privatization of data and secure data aggregation. In IoT networks, for example, studies have shown that FL is effective for real-time traffic prediction and anomaly detection. [7] This highlights FL's ability to maintain this dynamic input from numerous sources offering safe, scalable decision-making.

Despite these advances, there are still concerns with federated learning in IoT, mostly related to model accuracy, efficiency of communication, and safety concerns including the potential of model poisoning and data inference attacks. Through cryptographic protocols and reinforcement learning techniques, ongoing research seeks to strengthen the robustness of FL frameworks, improving model resilience and adaptation to real-time cyber threats [8]

We consider in this research proposal to enhance FL's security in IoT environments by ensuring both data integrity and confidentiality. For that, we propose integrating **HMAC-SHA256** [9] for verifying model update integrity and semi-Homomorphic encryption for secure data aggregation.

II. Literature Review

Federated learning has emerged as a prominent method for privacy-preserving collaborative ML in a decentralized fashion. Various frameworks have been proposed to improve the FL's privacy, efficiency, and adaptability in IoT environments. Below, are reviewed prior work relating to methods that focus on tackling the challenges in confidentiality, integrity, and communication efficiency.

One method that has surfaced for privacy-preserving collaborative machine learning (ML) is federated learning (FL), which harbors a decentralized approach. There have been multiple frameworks suggested as approaches to improve FL's privacy, efficiency, and adaptability in IoT environments. Previous literature was centered around addressing challenges related to confidentiality, integrity, and communication efficiency.

Fully Homomorphic Encryption (FHE) is one of the approaches that undertakes the challenge of data confidentiality via preventing unauthorized access to sensitive information exchanged between IoT devices and central servers during the FL process. Without the exposure of raw data, FHE allows encrypted data to be aggregated without the need of decryption. By way of data encryption throughout the computation process, FHE ensures the safety of the system by blocking access to attackers or inferring sensitive information from model parameters. While in some areas FHE may be efficient, it lacks in others. It is vulnerable to model poisoning and other data manipulation attacks during transmissions; as on the one hand it effectively protects confidentiality, on the other, it lacks mechanisms for ensuring data integrity. Furthermore, it is inappropriate for IoT devices which have limited processing power and memory because it introduces significant computational and communication overhead. Even though clustering may combat the issue of direct communication, it adds a layer of complexity to cluster management inevitably limiting scalability [10].

Secure Multiparty computation (SMPC) stands as an alternative approach as it addresses the issue of sensitive information exposure from transmitted model updates. Adversaries could still infer data from model updates even though FL stores local data. SMPC utilizes a secure aggregation protocol, ultimately allowing model updates from each node to be aggregated without revealing individual updates to the server. Consequently, this framework enhances privacy via masking each node's update, providing exposure only to the aggregated model [11]. Client dropout is also mitigated with secure aggregation as it is common in distributed IoT networks. SMPC, regardless of its benefits, also has several drawbacks. It lacks explicit integrity verification mechanisms, introduces significant computational and communication overhead due to the masking and aggregation processes, and does not offer adaptive protection against real time threats.

A dual approach encompassing both Homomorphic encryption (HE) and Differential Privacy (DP) [12] has also been explored to address privacy concerns. HE encrypts model updates thus eliminating the need for decryption at the server, enabling secure aggregation, while DP masks individual contributions and protects data privacy by adding noise to model parameters. However, the absence of mechanisms for verifying data integrity poses a limitation. Furthermore, HE imposes significant encryption costs, and the added noise from DP further impacts efficiency. Mechanisms for real-time threat detection and active attack prevention also seem to be blind spots when it comes to this approach.

While previous work provides robust solutions to ensure data privacy in FL, most seem to lack the mechanisms for ensuring the integrity of the transferred updates, have high computational costs, and are limited in real-time threat response. The proposed work aims to address these issues through combining Homomorphic Encryption and HMAC-SHA256.

1) **Homomorphic Encryption with TenSEAL**: Homomorphic Encryption is widely used in federated learning to enable computations on encrypted data, ensuring confidentiality throughout the training process. However, traditional implementations of Homomorphic Encryption are computationally expensive and impractical for resource-constrained IoT devices due to their high overhead. To address this challenge, **TenSEAL** - a Python library built on top of Microsoft SEAL - which enhances the use of Homomorphic Encryption is utilized. TenSEAL supports schemes like CKKS (for approximate arithmetic on floating-point numbers) and BFV (for exact operations on integers), both of which are well-suited for federated learning tasks.
   TenSEAL enables practical operations, such as addition and multiplication, on encrypted data, which are sufficient for securely aggregating model updates in FL. By offering and leveraging optimized implementations from SEAL, TenSEAL reduces the computational complexity typically associated with HE while maintaining a balance between security and performance. Making TenSEAL a practical choice for IoT environments, where nodes would have limited processing power and memory. As a result, TenSEAL ensures that model updates remain confidential without imposing significant resource constraints, enabling secure and efficient federated learning in real-world scenarios.

2) **HMAC-SHA256 for Integrity Verification**: While federated learning frameworks often focus on data confidentiality, they frequently overlook integrity verification, leaving systems vulnerable to tampering and model poisoning attacks. To address this, we use **HMAC-SHA256**, a cryptographic hashing mechanism that ensures data integrity with minimal computational overhead. HMAC-SHA256 generates a unique hash for each model update, enabling the receiver to verify that the transmitted data has not been altered during transmission. Its lightweight design and fast computation make it particularly well-suited for IoT devices, which have limited resources. By incorporating HMAC-SHA256 into the FL workflow, the proposed framework provides robust integrity verification without compromising performance. This ensures the reliability and security of the aggregated model, as any tampering or active attacks can be immediately detected and mitigated.

The dual-layer approach that combines both Homomorphic Encryption for confidentiality and HMAC-SHA256 for integrity, addresses key challenges in IoT-based Federated Learning; this includes resource constraints, communication overhead, and real-time threat detection. Different from previous models that centralize their approach on confidentiality,

this model endorses both data privacy and integrity verification, creating a well-suited model for IoT environments.

The framework decreases computational complexity while preserving efficient confidentiality which is ideal for resource limited IoT devices by utilizing semi-Homomorphic encryption (SHE). To add on, real-time detection of tampering and active threats can be achieved through HMAC-SHE256 as it provides lightweight and effective integrity checks. This dual-layered approach is set aside from more resource-intensive solutions like Fully Homomorphic Encryption (FHE) or Secure Multiparty Computation (SMPC) because it achieves a balance between security and performance.

Although previous methods address privacy concerns, they often overlook computational efficiency, data integrity, and real-time threat resistance. FHE imposes high computational costs and lacks even though it ensures strong privacy. SMPC also proves inherently flawed as it suffers from significant overhead and offers no tamper protection. CKKS encryption enhances computational efficiency but remains vulnerable to model tampering. Similarly, the merging of HE with Differential Privacy (DP) protects privacy but does not verify data integrity. The proposed solution overcomes these limitations, offering a practical and secure framework for IoT-based FL.

TABLE I
COMPARISON OF METHODS AND FRAMEWORKS FOR FEDERATED LEARNING IN IOT - CONFIDENTIALITY AND INTEGRITY

Method/Framework	Data Confidentiality	Data Integ
Fully Homomorphic Encryption (FHE)	Yes	No
Secure Multiparty Computation (SMPC)	Yes	No
CKKS Scheme of Homomorphic Encryption	Yes	No
Homomorphic Encryption (HE) + Differential Privacy (DP)	Yes	No
semi-Homomorphic Encryption (SHE) + HMAC-SHA256	Yes	Limited (through HMAC-SHA256)

TABLE II
COMPARISON OF METHODS AND FRAMEWORKS FOR FEDERATED LEARNING IN IOT - COMPUTATIONAL EFFICIENCY

Method/Framework	Computational Efficiency
HE	Low (due to high computational cost)
SMPC	Moderate (high due to masking/aggregation)
CKKS Scheme of Homomorphic Encryption	Low (high latency and resource requirements)
HE + DP	Moderate (additional noise injection impacts efficiency)
Optimized HE using TenSEAL + HMAC-SHA256	High (optimized and computationally efficient for IoT)

TABLE III
COMPARISON OF METHODS AND FRAMEWORKS FOR FEDERATED LEARNING IN IOT - COMMUNICATION OVERHEAD & PROBLEMS ADDRESSED

Method/Framework	Communication Overhead & Problems Addressed
HE	High, but effective for confidentiality. No integrity verification, high computational cost unsuitable for IoT.
SMPC	High, lacks real-time threat response and integrity verification. Effective in confidentiality but limited in adaptive threat protection.
CKKS Scheme of Homomorphic Encryption	High, lacks data integrity checks and real-time threat mitigation. Suitable for privacy but computationally demanding.
HE + DP	High, lacks real-time threat detection and active attack prevention. Good for privacy, but lacks data integrity verification.
Optimized HE using TenSEAL + HMAC-SHA256	Moderate, addresses computational efficiency for IoT, maintains data integrity & confidentiality with reduced overhead and computational cost.

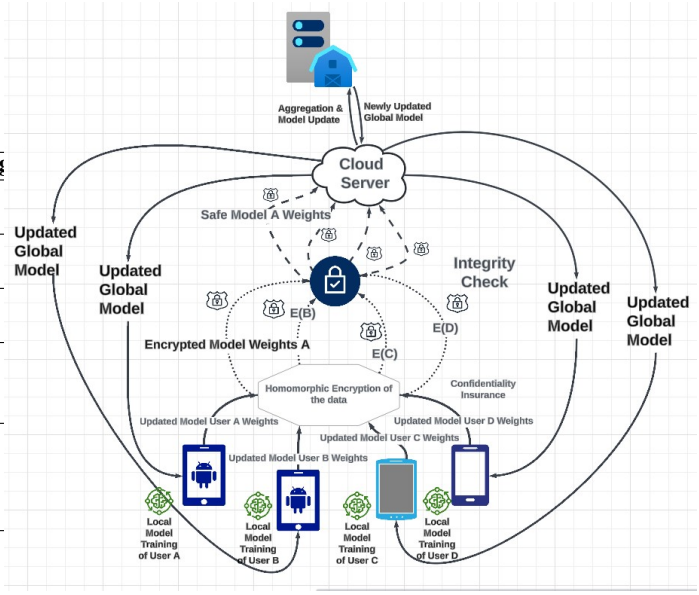

Fig. 1. Network composed of edge devices exchanging model weights securely to the central server through Federated Learning.

III. SYSTEM MODEL

As shown in Figure 1, we consider an IoT environment where multiple edge devices collaborate to train a shared global model. Each device collects data locally, trains a model using this data, and performs local updates. These updates are then securely aggregated to ensure privacy while preserving model accuracy.

A. Edge Devices (Client Nodes)

In this framework, a global model is distributed across IoT devices (e.g., in Figure 1, User A, B, C, D) where these nodes

train the allocated model individually using their allocated local data.

B. Encryption and adding the HMAC

Once training finishes the node returns a set of the model's coefficients and intercept that are encrypted using homomorphic encryption, ensuring data confidentiality. An HMAC signature is generated ensures the data's integrity during transmission. Finally, the encrypted weights and their signature are stored for secure use in federated learning or similar decentralized systems.

C. Aggregation and Transfer of Data

The encrypted weights and intercepts from all clients are added, then we scale the total by the reciprocal of the number of clients to compute the average. This process maintains the confidentiality of the data, as the averaging is performed directly on encrypted values.

After aggregation, the resulting values are then decrypted, forwarded and later update the global model's parameters. Once a new global model is computed, this model is then sent to each individual client and back to the step of Edge Devices.

IV. Proposed Approach

This study implements a federated learning framework secured with HE using **TenSEAL** for data confidentiality and **HMAC-SHA256** for integrity verification. The proposed framework focuses on IoT environments where data privacy and resource constraints are critical considerations.

We begin by pre-processing the CDC Diabetes Health Indicators dataset. Correlation analysis is performed to identify strong relationships between variables and the unbalanced data is addressed using **SMOTENC**, which generates synthetic samples to balance the classes. After preprocessing, a pre-trained model from **TensorFlow Keras** serves as the base global model.

For simulation purposes, edge devices are represented as isolated Python environments to emulate multiple IoT devices. Each device trains the local model using its preprocessed data subset.

Once training is complete, the model updates are encrypted using **TenSEAL**, a library that implements Fully Homomorphic Encryption, ensuring that sensitive information remains protected. Simultaneously, **HMAC-SHA256** hashes are generated to verify the integrity of the updates.

The initialized global model which would be on the central server would receive the updated, aggregated weights and update the model to be redistributed.

To evaluate the proposed framework, experiments are conducted to measure its resilience against both passive and active attacks. Metrics such as data leakage, tamper detection rates, and the impact of model poisoning attacks on global model accuracy are assessed. Additionally, the computational and communication overhead introduced by encryption and integrity verification are analyzed, and the results are compared against a baseline federated learning framework without security enhancements.

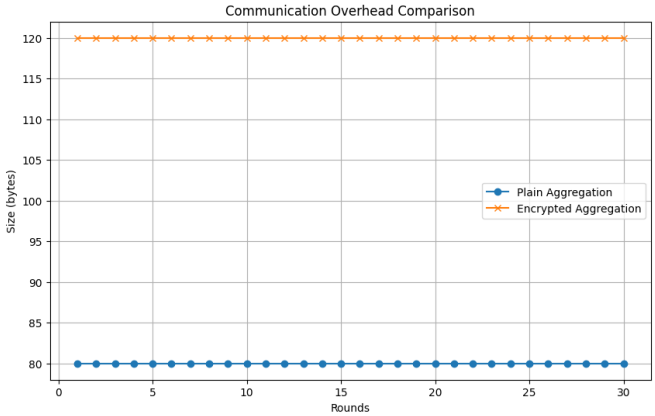

Fig. 2. Communication overhead comparison between plain and encrypted aggregation over 30 rounds.

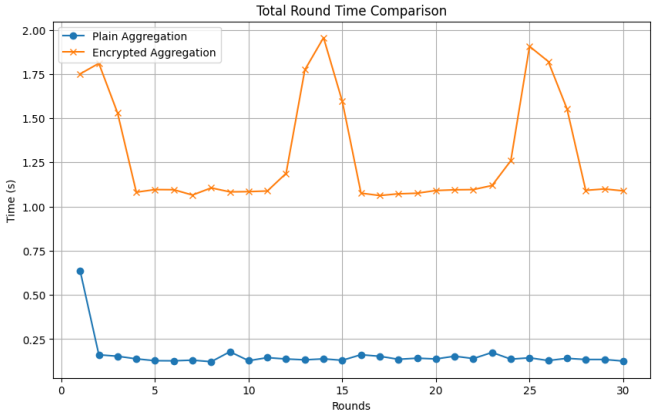

Fig. 3. Total round time comparison showing the computational impact of encryption.

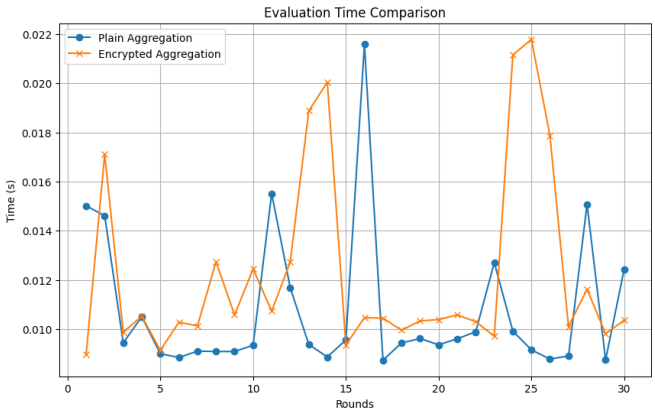

Fig. 4. Evaluation time comparison between plain and encrypted aggregation.

V. Visualization of Experimental Setup

VI. Results & Analysis

A. Communication Overhead

The encrypted environment introduced a 50% increase in communication overhead compared to plain aggregation. Plain

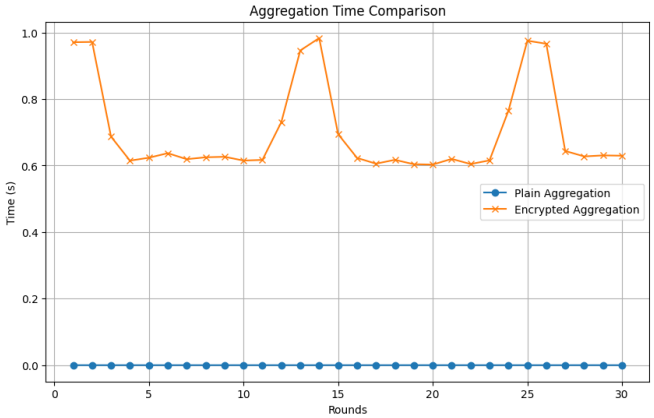

Fig. 5. Aggregation time comparison highlighting the computational burden of encryption.

aggregation consistently required approximately 80 bytes per round, whereas encrypted aggregation incurred an overhead of approximately 120 bytes per round. This increase is attributed to the additional cryptographic metadata introduced during encryption.

B. Round Time and Aggregation Time

The total round time for encrypted aggregation ranged between 1.0 and 2.0 seconds, significantly exceeding the range for plain aggregation ($\approx$ 0.1–0.2 seconds). Aggregation time in the encrypted setting peaked at approximately 1.0 seconds, while plain aggregation remained negligible ($\approx$ 0 seconds). These results underscore the computational burden imposed by homomorphic encryption, particularly during aggregation, identifying it as a key performance bottleneck in federated learning (FL).

C. Training and Evaluation Time

Training times in the encrypted environment ranged from $\approx$ 0.4 to 0.9 seconds, compared to plain aggregation's stable range of $\approx$ 0.1 to 0.2 seconds. Despite the additional overhead, evaluation times for both settings remained consistently low ($\approx$ 0.01–0.02 seconds), indicating minimal impact of encryption on the evaluation process.

D. Accuracy and Loss

Global accuracy and client loss were unaffected by encryption. Both aggregation methods achieved an overall global accuracy of 75.75% across 30 FL rounds. Similarly, the average client loss remained stable, ranging between 0.493 and 0.498, demonstrating that the encryption process does not compromise model performance.

References

[1] IBM Research, "What is federated learning?" 2023, [Online; accessed November 11, 2024]. [Online]. Available: https://research.ibm.com/blog/what-is-federated-learning

[2] Q. Yang, Y. Liu, T. Chen, and Y. Tong, "Federated machine learning: Concept and applications," *ACM Transactions on Intelligent Systems and Technology*, vol. 10, no. 2, pp. 12:1–12:19, 2019. [Online]. Available: https://doi.org/10.1145/3298981

[3] Statista Research Department, "Number of connected iot devices worldwide from 2019 to 2030," 2024, [Online; accessed November 11, 2024]. [Online]. Available: https://www.statista.com/statistics/1101442/iot-number-of-connected-devices-worldwide/

[4] M. Beck, "How much data is the iot generating?" 2018, [Online; accessed November 11, 2024]. [Online]. Available: https://www.sumologic.com/blog/iot-data-volume

[5] O. J. Ademolu Ajao and E. Adetiba, "The applications of federated learning algorithm in the federated cloud environment: A systematic review," in *International Conference on Science, Engineering and Business for Driving Sustainable Development Goals (SEB4SDG)*. IEEE, 2024. [Online]. Available: https://doi.org/10.1109/SEB4SDG60871.2024.10629812

[6] M. Kumar, A. Singh, and B. Gupta, "A survey on federated learning in iot: Challenges and applications," *International Journal of Machine Learning and Cybernetics*, vol. 13, pp. 1823–1841, 2022. [Online]. Available: https://link.springer.com/article/10.1007/s13042-022-01647-y

[7] S. R. A. K. S. Arman Raj, Vandana Sharma and N. Kumar, "Strengthening the security of iot devices through federated learning: A comprehensive study," *11th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO)*, 2024.

[8] P. N. M. Shreyas R. Dixit and G. R. Shinde, "Rethinking data integrity in federated learning: Are we ready?" in *IEEE International Women in Engineering Conference on Electrical and Computer Engineering (WIECON-ECE)*. Pune, India: IEEE, 2022.

[9] Wikipedia contributors, "Hmac – wikipedia, the free encyclopedia," 2024, [Online; accessed November 11, 2024]. [Online]. Available: https://en.wikipedia.org/wiki/HMAC

[10] M. G. B. O. N. M. Hijazi, M. Aloqaily and F. Karray, "Secure federated learning with fully homomorphic encryption for iot communications," *IEEE Internet of Things Journal*, vol. 11, no. 3, pp. 4289–4304, 2024.

[11] C. L. Sogo Pierre Sanon, Rekha Reddy and H. D. Schotten, "Secure federated learning: An evaluation of homomorphic encrypted network traffic prediction," in *IEEE Consumer Communications and Networking Conference (CCNC)*. Las Vegas, NV, USA: IEEE, 2023.

[12] O. S. R. S. Arnaud Grivet Sebert, Marina Checri and C. Gouy-Pailler, "Combining homomorphic encryption and differential privacy in federated learning," in *2023 20th Annual International Conference on Privacy, Security and Trust (PST)*, Copenhagen, Denmark, 2023, pp. 1–7. [Online]. Available: https://ieeexplore.ieee.org/document/10320195